

SNEHAN BISWAS

Senior Consultant – Generative AI | RAG & Agentic AI Systems

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PROFESSIONAL SUMMARY

Senior Consultant in Generative AI at Capgemini India with over five years of hands-on experience in designing, building and deploying generative AI applications, with a strong specialisation in Python-based implementations and cloud services integration. Has led and contributed to Retrieval-Augmented Generation (RAG) initiatives for international clients, including a RAG-based proof of concept for a communications client built on AWS services and Google Gemini models, and Video-RAG and codebase-aware RAG systems for a US-based semiconductor chip design manufacturer using Azure OpenAI and Azure AI Search. Currently supporting the development and deployment of multi-agent AI applications orchestrated with LangGraph and the Google A2A protocol.

Previously a Senior Systems Analyst at IEMA Research and Development Pvt. Ltd., where he engineered the organisation's end-to-end software system IEMAIWriter and evolved it into the **iema.ai** platform, while delivering AI and ML solutions to clients such as the Indian Army, SAIL and Ericsson. Brings deep expertise in backend API development, multi-threaded application design and data privacy compliance, combined with a sustained commitment to emerging-technology research and cross-functional collaboration.

CORE COMPETENCIES

Generative AI	Generative AI application development, AI solution design, Retrieval-Augmented Generation (RAG), Adaptive RAG, Video-RAG, multi-agent systems with the Google A2A protocol
Frameworks & Libraries	LangChain, LangGraph, FastAPI, Streamlit, Hugging Face, Ollama, OpenAI
Vector Databases	ChromaDB, FAISS, Azure AI Search, ensemble vector stores with weighted mapping
Microsoft Azure	Azure OpenAI, Azure AI Search, Azure-based inference pipelines
Amazon Web Services	AWS Bedrock, AWS Lambda, AWS S3, AWS DynamoDB, AWS EC2
Software Engineering	Python programming, backend API development, multi-threaded application development, offline AI applications, automation scripting (Python, PowerShell)
Professional	Data protection and privacy compliance, requirements analysis, system documentation, cross-functional collaboration, technical mentoring

PROFESSIONAL EXPERIENCE

Capgemini India

Kolkata, India

Senior Consultant – Generative AI

Jun 2026 – Present

Generative AI Consultant

Mar 2025 – Jun 2026

Developing generative AI applications for global clients, with a focus on Python implementations that apply a broad range of generative AI concepts, from retrieval-augmented generation to agentic workflows.

- **RAG-based PoC for a communications client:** Designed a retrieval-augmented generation proof of concept grounded in AWS service documentation, powered by the Google Gemini model and built with LangChain and FastAPI. The system used a custom knowledge base constructed on top of AWS services including Lambda, S3, DynamoDB and Bedrock.
- **Video-RAG system for a semiconductor client:** Collaborated with a US-based semiconductor chip design manufacturer to build a Video-RAG system that allows users to interact with video data through retrieval-augmented generation, grounded in Azure LLM models via Azure OpenAI and Azure AI Search.
- **Codebase-aware Adaptive RAG:** Recognising that a project's codebase is the most important resource for understanding it, designed and developed a PoC for the same client that functions as an adaptive RAG over a vector store built from the entire project codebase, enabling faster development. The inference pipeline was built primarily on Azure services and delivered to users through a Streamlit interface.
- **Production GenAI chatbot:** Worked on a production generative AI chatbot that the client's employees use for Travel and Finance related queries, contributing to both the design and deployment phases. The application uses a ReactJS frontend and a Python backend built with LangChain and LangGraph, with workflows exposed through FastAPI.

- **Multi-agent AI application:** Working with a US-based communications client, supporting a combined offshore and onshore team in the development and deployment of a multi-agentic AI application that uses the Google A2A protocol for agent communication and LangGraph for agent workflow orchestration.
- **Applied research:** Conducted research at the Capgemini Kolkata Research Desk in collaboration with the University of Calcutta, designing a module for efficient RAG application development based on weighted mapping across ensemble vector stores.

IEMA Research and Development Pvt. Ltd.

Kolkata, India

Senior Systems Analyst

Sep 2021 – Mar 2025

Delivered solutions in the domain of artificial intelligence and machine learning for a diverse client base including the Indian Army, SAIL, Ericsson and the IEM-UEM Group.

Product Engineering

- Engineered the organisation's comprehensive end-to-end software system, **IEMAIWriter**, and led its evolution into the **iema.ai** platform ([iema.ai](#)) to strengthen organisational capabilities.
- Developed the backend of **IEMAI-Tutor**, the AI tutoring module incorporated within iema.ai.
- Created end-to-end offline AI applications using pure Python frameworks.

Systems Analysis & Delivery

- Collaborated with cross-functional teams to develop innovative solutions for complex problems.
- Defined requirements for new or modified systems in close collaboration with stakeholders, ensuring alignment with organisational goals.
- Analysed existing systems to identify potential problems and areas for improvement, and identified opportunities for process optimisation through technology.
- Developed, tested and implemented system changes to ensure the successful rollout of new systems, and documented system designs, configurations and test results according to established standards.
- Developed custom scripts in Python and PowerShell to automate routine tasks.

Governance, Research & Leadership

- Ensured compliance with applicable laws and regulations related to data privacy.
- Researched emerging technologies to assess their applicability within the organisation's environment.
- Provided guidance and mentorship to junior team members on technical matters.

EDUCATION

Bachelor of Technology – Electrical, Electronics and Communications Engineering

Jun 2021

University of Engineering and Management, Kolkata, West Bengal

Indian School Certificate (ISC) – Physics, Chemistry, Mathematics, Computer Science

Jun 2017

The Modern Academy, Kolkata

Indian Certificate of Secondary Education (ICSE) – Science

Jun 2015

Ram Mohan Mission High School, Kolkata

HONOURS & AWARDS

- **Vice-Chancellor's Award for Best Techno Wiz** – University of Engineering and Management, recognised across the 2017–2021 academic period.
- **First Prize for Internal Performance** – Indian School Certificate (ISC) Examination, 2017.

LANGUAGES

English C2 – Proficient	Bengali C2 – Proficient	Hindi C1 – Advanced
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ACTIVITIES & INTERESTS

- **Teaching:** Provides tuition to second- and third-year undergraduate students in Electronics and Communications Engineering.
- **Interests:** Exploring new places on motorbike and car tours, playing the guitar, collecting rare currencies, listening to classic songs, and coding over a cup of coffee.